

Total No. of Questions : 8]

PD4697

SEAT No. :

[Total No. of Pages : 4

[6404]-203

B.E.(Mechanical Engineering)

**HEATING VENTILATION AIR-CONDITIONING AND
REFRIGERATION**

(2019 Pattern) (Semester - VII) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket Calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Discuss any one method of capacity controls of centrifugal compressor. **[6]**

b) Explain with neat sketch thermostatic expansion valve. **[6]**

c) Explain oil pressure failure control. **[5]**

OR

Q2) a) Explain with neat sketch evaporative condenser. **[6]**

b) Explain with neat sketch low temperature control. **[6]**

c) Explain with neat sketch transcritical cycle. **[5]**

Q3) a) Discuss the various types of indoor air contaminants. **[8]**

b) Air at 25°C and 60% RH is supplied to the drug formulation unit. It is conditioned to this state, first by cooling and dehumidification and then by reheating it. Cooling coil surface temperature is 13°C and ambient conditions are at 32°C DBT & 65% RH. if the supply air rate is 15,000 m³/hr, determine: **[10]**

- i) Cooling coil capacity in TR
- ii) BPF of coil
- iii) Heating capacity in kW
- iv) Heating coil surface temperature if BPF is 0.3
- v) Mass of water removed

OR

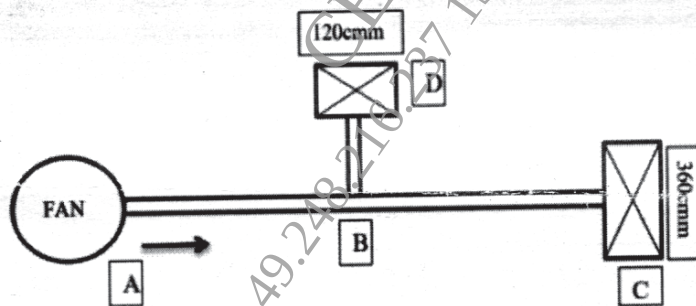
P.T.O.

- Q4) a)** What is CLTD method? How it connects with time lag and decrement factor. [8]
- b)** Outside design conditions are 40°C and 30% RH. Room design conditions are 25°C and 50% RH. Room sensible heat is 50 kW and room latent heat is 10 kW. If the outside air quantity is $50\text{m}^3/\text{min}$ and assuming by pass factor of cooling coil is 0.1. Find: GSHF and ESHF. [10]

- Q5) a)** Explain selection and location of supply outlets. [8]
- b)** A circular duct of 40 cm is selected to carry air at velocity 440 meter per minute if duct is replaced by rectangular duct of aspect ratio 1.5. Find the size of a rectangular duct for equal friction when;
- Velocity in two duct is same.
 - Discharge in two duct is same.
- If $f = 0.015$. Find the pressure loss per 100 m length of duct. Take density of air = 1.15 kg/m^3 . [10]

OR

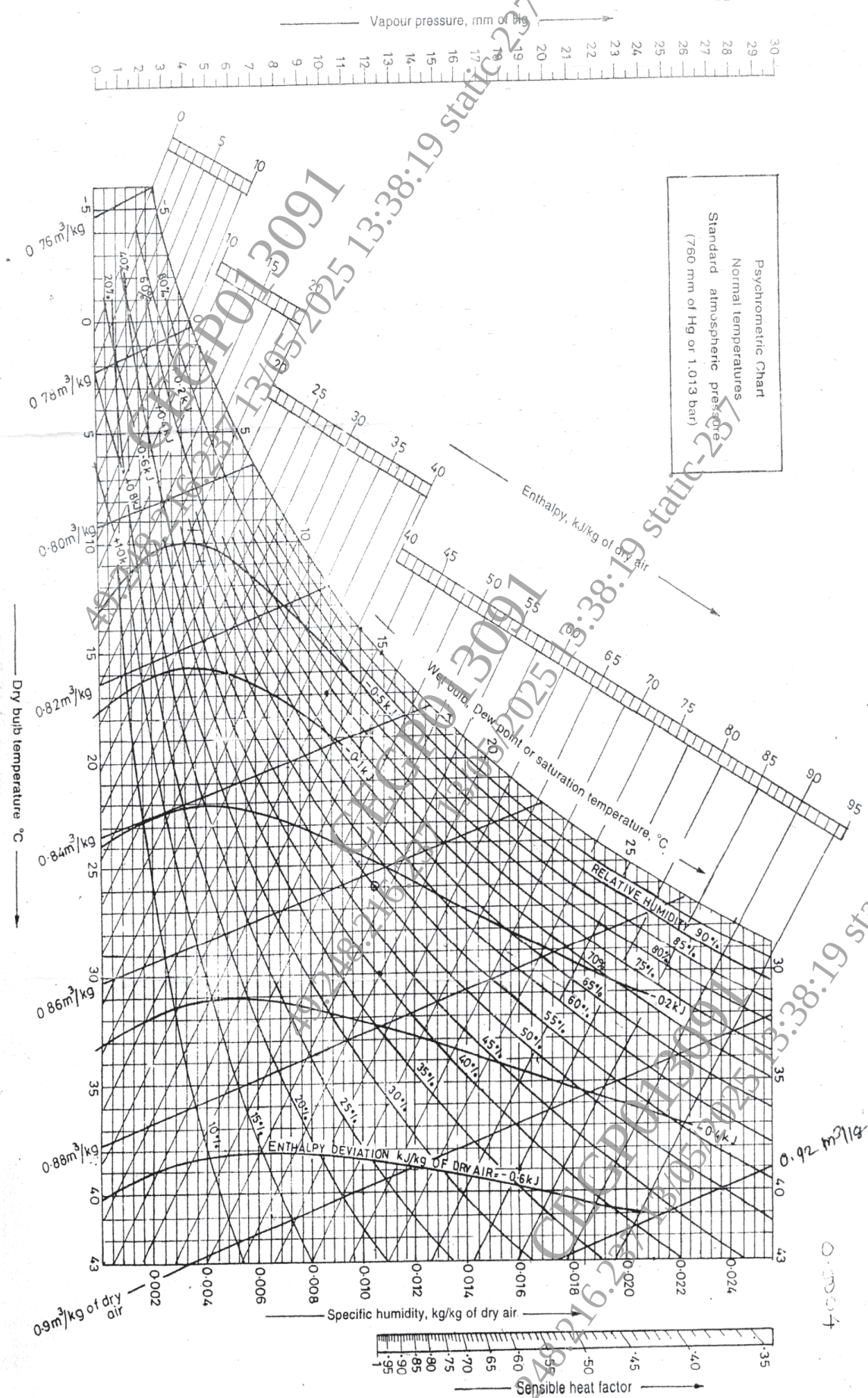
- Q6) a)** Explain Natural & Mechanical ventilation. [8]
- b)** Using Equal friction method, determine the duct diameter and velocity for section AB, BD and BC. Assume velocity in the main duct AB = 600 m/min. Also Calculate maximum pressure drop in the duct system. Distance AB = 30m, Distance BC = 30m and Distance BD = 10m. Refer the figure as given below. [10]

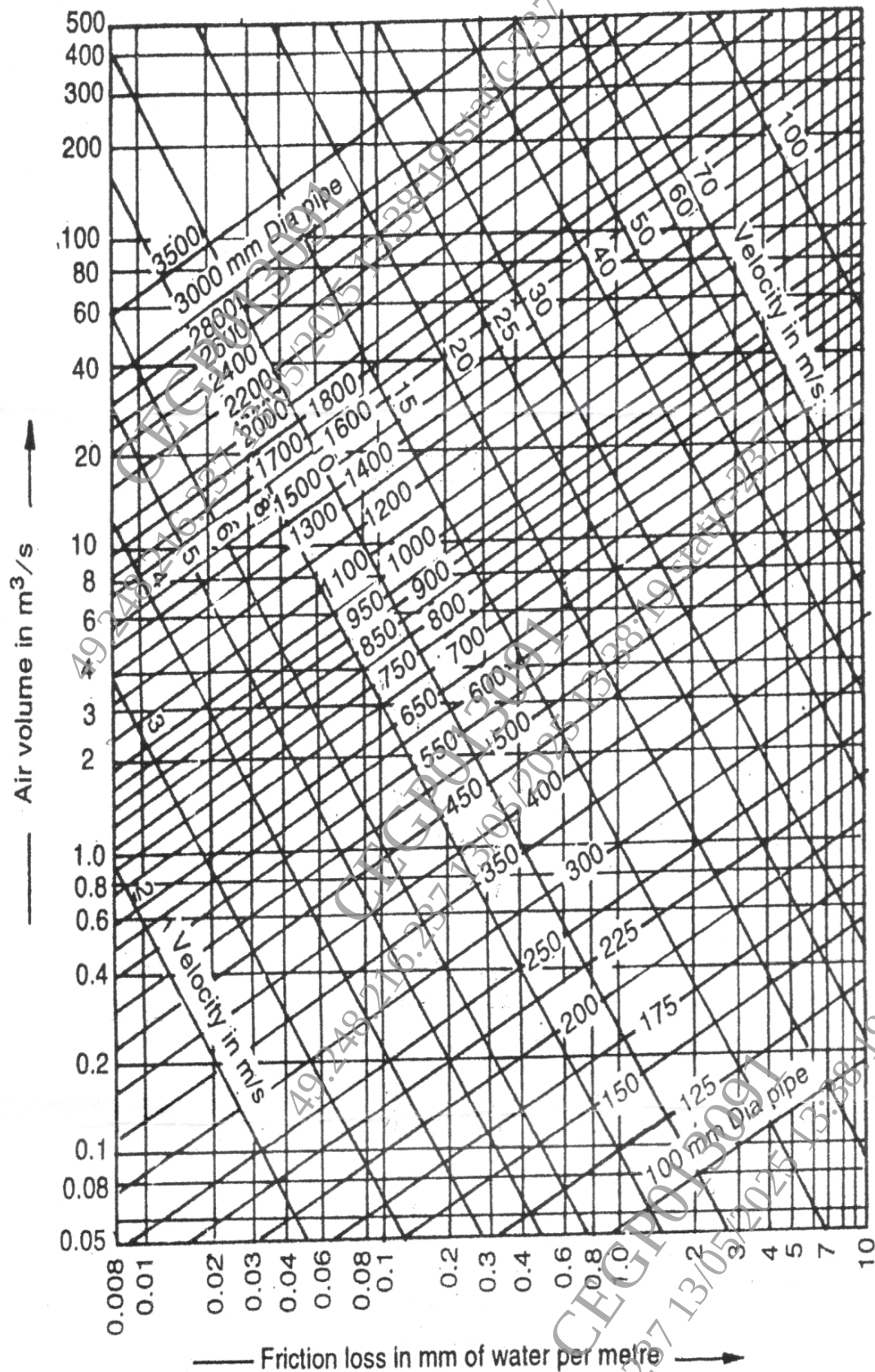


- Q7) a)** Draw neat sketch and explain all year round AC system. [6]
- b)** Draw neat sketch and explain solid desiccant air conditioning system. [6]
- c)** Explain evaporative cooling air conditioning systems. [5]

OR

- Q8) a)** Explain with neat sketch all-water system. [6]
- b)** Explain with neat sketch solid packed tower. [6]
- c)** Explain clean room air conditioning system. [5]





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